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O. A. Hurricane,^{a)} D. T. Casey, O. Landen, A. L. Kritcher, R. Nora, P. K. Patel, J. A. Gaffney, K. D. Humbird, J. E. Field, M. K. G. Kruse, J. L. Peterson, and B. K. Spears

AFFILIATIONS

Lawrence Livermore National Laboratory, P.O. Box 808, L-472, Livermore, California 94550, USA

^{a)}Author to whom correspondence should be addressed: hurricane1@llnl.gov

ABSTRACT

For many years, low mode asymmetry in inertially confined fusion (ICF) implosions has been recognized as a potential performance limiting factor, but analysis has been limited to using simulations and searching for data correlations. Herein, an analytically solvable model based upon the simple picture of an asymmetric piston is presented. Asymmetry of the shell driving the implosion, as opposed to asymmetry in the hot-spot, is key to the model. The model provides a unifying framework for the action of mode-1 shell asymmetry and the resulting connections between various diagnostic signatures. A key variable in the model is the shell asymmetry fraction, f , which is related to the areal density variation of the shell surrounding the hot-spot. It is shown that f is simply related to the observed hot-spot mode-1 velocity and to the concept of residual energy in an implosion. The model presented in this paper yields explicit expressions for the hot-spot diameter, stagnation pressure, hot-spot energy, inertial confinement-time, Lawson parameter, hot-spot temperature, and fusion yield under the action of mode-1 asymmetry. Agreement is found between the theory scalings when compared to ICF implosion data from the National Ignition Facility and to large ensembles of detailed simulations, making the theory a useful tool for interpreting data. The theory provides a basis for setting tolerable limits on asymmetry.

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I. INTRODUCTION

Unfortunately, there are numerous ways to degrade the performance of an inertially confined fusion (ICF) implosion. Few question that instability and mix are bad for ICF implosions. However, experimental data on the National Ignition Facility (NIF) rarely demonstrate any clear correlations of performance with the hot-spot shape. Nevertheless, simulations and numerical models repeatedly show low mode asymmetry and the resulting residual kinetic energy (RKE) causes degradation.^{1–9} Confusion on the topic of asymmetry in ICF implosions has been made worse by the conflation of the easily observable, through emission hot-spot asymmetry with more difficulty to directly observe asymmetry of the cold dense shell surrounding the hot-spot.

National Ignition Facility (NIF) ICF implosions are commonly observed to exhibit asymmetric hot-spots as directly seen in x-ray emission and neutron emission. There is also the indirect indication of hot-spot asymmetry, in the form of flows in the neutron producing regions that are inferred from neutron time-of-flight (NToF)

diagnostics along different lines-of-sight^{10,11} and in the different burn-average Brysk temperatures measured between deuterium-tritium (DT) and deuterium-deuterium (DD) reactions.^{12–14} Yet, there is little clear correlation between the observed hot-spot asymmetry and DT implosion yield performance in the vast majority of the NIF database, albeit a handful of sub-sets of implosion that show yield vs asymmetry correlation can be found⁹ but whether the correlations are convincing is arguable.

There is an apparent correlation between hot-spot mode-1 velocity (v_{p1}) and nuclear measurements that characterize the shell of the implosion.¹⁵ Asymmetry in the areal density ($\rho\delta R$, where ρ is density and δR is thickness) of the cold shell, defined as the DT main fuel layer and remaining ablator material, can be inferred from flange nuclear activation diagnostics (FNAD) and neutron time-of-flight (NToF) diagnostics along different lines-of-sight. Empirically, the NToF measured v_{p1} does correlate with the degradation of fusion yield in gas-capsule implosions¹¹ and in layered DT implosions,¹⁵ but a theoretical framework has been lacking.

Herein, a simple analytical model is described that captures the essential element of mode-1 asymmetry in an ICF implosion. The model provides an integrating framework for the action of mode-1 shell asymmetry on fusion performance and on various diagnostic signatures. The model gives a clear explanation for the observed hot-spot mode-1 velocity and gives the connection between this mode-1 velocity, implosion velocity, shell areal density asymmetry, and RKE. The model presented in this paper yields explicit expressions for the hot-spot diameter (S_{min}), stagnation pressure (P_{stag}), hot-spot energy at stagnation (E_{hs}), inertial confinement-time (τ), Lawson¹⁶ parameter ($P\tau$), hot-spot thermal temperature (T_{hs}), and fusion yield (Y) under the action of mode-1 asymmetry. In all cases, the model shows that degradation of these quantities is related to one elementary parameter, f , the “asymmetry factor.” Finally, the scalings coming from the theory are compared to a suite of ensemble simulations and directly to DT implosion data from NIF where agreement, of the quantities compared, is found in both cases.

II. ANALYTIC IMPLOSION MODEL FOR MODE-1 ASYMMETRY

A simple yet surprisingly powerful analytical model for mode-1 asymmetry in an ICF implosion can be constructed by abstracting a three-dimensional implosion decelerating from peak velocity, into an asymmetric piston, of cross-sectional area A , as shown in Fig. 1. The asymmetric pistons, which represent segments of the implosion shell, are characterized by a minimum and maximum areal density ($\rho\delta R_{min}$ and $\rho\delta R_{max}$). In the development below, it is assumed that the pistons are incompressible and constant mass, which clearly neglects some of the physics details of a real ICF implosion shell as it stagnates on the hot-spot, but as will be seen below this assumption appears to be adequate for back-of-the-envelope predictive capability.

The asymmetric pistons are separated by a “hot-spot” plasma with time (t) varying plasma pressure $P(t)$ and the hot-spot-side of each piston is located at different spatial coordinates, $x_1(t)$ and $x_2(t)$, respectively (see Fig. 1). The equations of motion for each piston are given by

$$\rho\delta R_{min}\ddot{x}_1 = -P, \quad (1)$$

$$\rho\delta R_{max}\ddot{x}_2 = P, \quad (2)$$

where over-dots indicated time-derivatives. The separation between the pistons, $S(t) = x_2(t) - x_1(t)$, defines the hot-spot diameter. As a result of the way, an ICF hot-spot is formed; for low levels of alpha-heating, the thermodynamics of an ICF hot-spot are nearly adiabatic,¹⁷ $PV^\gamma \sim \text{const.}$, where $V(t) = AS(t)$ is the hot-spot volume (based upon the abstraction shown in Fig. 1) and γ is the polytropic index ($\gamma = 5/3$ for an ideal gas). So, the hot-spot pressure can be related to the piston separation

$$P(t) = P_0 \left(\frac{S_0}{S(t)} \right)^{5/3}, \quad (3)$$

where $P_0 = P(0)$ and $S_0 = S(0)$ are understood to be the hot-spot pressure and piston separation at peak implosion velocity, respectively.

Subtracting Eq. (1) from Eq. (2) and using Eq. (3) yield an autonomous non-linear differential equation for the hot-spot diameter

$$\ddot{S} = \frac{P_0}{\rho\delta R_{ave}} \left(\frac{2}{1-f^2} \right) \left(\frac{S_0}{S} \right)^{5/3}, \quad (4)$$

where f , the shell asymmetry fraction, is defined as a measure of areal density variation from the mean shell areal density ($\rho\delta R_{ave}$)

$$\rho\delta R_{max} = \rho\delta R_{ave}(1+f), \quad (5)$$

$$\rho\delta R_{min} = \rho\delta R_{ave}(1-f), \quad (6)$$

or equivalently

$$f = \frac{\rho\delta R_{max} - \rho\delta R_{min}}{\rho\delta R_{max} + \rho\delta R_{min}}. \quad (7)$$

Note that the denominator of Eq. (7) can equivalently be written as $2\rho\delta R_{ave}$. As will be seen below, f is a key quantity that is simply related to the hot-spot mode-1 velocity and to the concept of residual kinetic energy (RKE). Since the asymmetric pistons were assumed to be

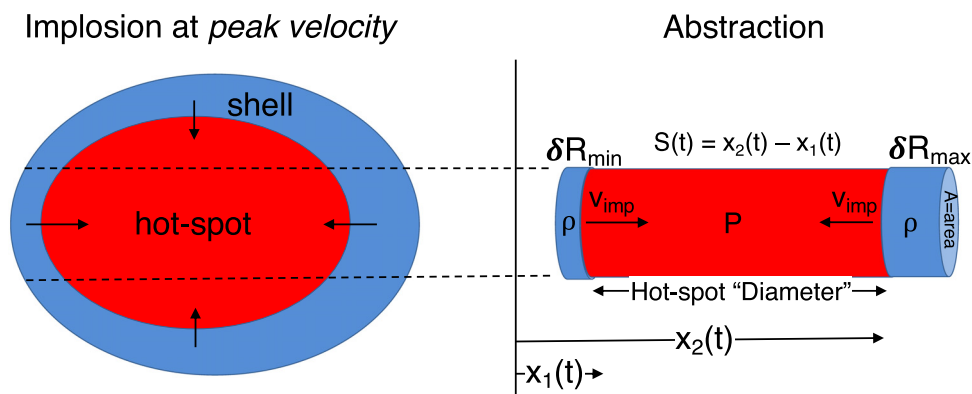


FIG. 1. (Left) An imploding shell is compressing a hotter, but lower density hot-spot as the shell decelerates from peak velocity. The shell and hot-spot are asymmetric in shape as pictured. Moreover, the shell has a thickness (areal density) variation. (Right) Along the direction defined by the extremes of shell thickness variation, the implosion is abstracted into an “apple core” of thin and thick pistons at opposing ends of the hot-spot. In this picture (right), the asymmetry of the hot-spot clearly has no meaning nor does the shape of the shell other than through the areal density difference of the opposing pistons. The three-dimensional (3D) orientation of the mode-1 asymmetry is arbitrary in this model.

incompressible and constant mass, f has no time-dependence so it will be treated as a constant in the subsequent integrations.

The first integral of Eq. (4) is easily accomplished by noting that $\dot{S} = \dot{S}(d\dot{S}/dS)$ and that there is always a constant speed frame-of-reference where the asymmetric pistons are initially moving toward each other at the same speed, v_{imp} , so that $\dot{S}(0) = -2v_{imp}$. Thus

$$\dot{S} = \pm \sqrt{4v_{imp}^2 + \frac{6P_0}{\rho\delta R_{ave}} \left(\frac{S_0}{1-f^2} \right) \left[1 - \left(\frac{S_0}{S} \right)^{2/3} \right]}, \quad (8)$$

where the sign of the expression on the right is determined by whether the pistons are imploding (−) or exploding (+). The integral of Eq. (8), which gives the full solution of Eq. (4) as an implicit equation for $S(t)$, is found to be after a little algebra

$$\sqrt{\left(\frac{S}{S_0} \right)^{2/3} - \frac{b}{1+b}} \left[\left(\frac{S}{S_0} \right)^{2/3} + \frac{2b}{1+b} \right] - \frac{1}{\sqrt{1+b}} \left(\frac{1+3b}{1+b} \right) = \pm \frac{2v_{imp}t\sqrt{1+b}}{S_0}. \quad (9)$$

In Eq. (9), $b = 3P_0S_0/[2\rho\delta R_{ave}v_{imp}^2(1-f^2)]$, which is normally a small parameter for an ICF implosion because $\rho v_{imp}^2 \gg P_0$. The expression above, Eq. (9), will not be needed elsewhere in this paper and is only included for completeness. However, note from Eq. (9) that the characteristic timescale from the right-hand-side of the equation, $S_0/(2v_{imp})$, is a deceleration-time that for a real implosion is related to the concept of “coast-time”¹⁸ and is also related to the rate of conversion of the kinetic energy of the implosion into pdV work on the hot-spot.

III. HOT-SPOT EQUATIONS

With Eq. (8), one can write down many key properties of the hot-spot at peak compression (i.e., minimal piston separation, S_{min}). Solving Eq. (8) for S with $\dot{S} = 0$, one obtains

$$S_{min} = \frac{S_0}{\left[1 + \frac{2\rho\delta R_{ave}v_{imp}^2}{P_0S_0}(1-f^2) \right]^{3/2}}. \quad (10)$$

Equation (10) relates the compression ratio, S_0/S_{min} , to the parameter b defined above. As will be seen below, $\rho\delta R_{ave}v_{imp}^2(1-f^2)$ in parameter b is a measure of hot-spot energy per unit area, A , at stagnation. Note that upon substituting Eq. (10) back into Eq. (9) gives a time at which stagnation occurs, $t(S_{min})$, that numerically is very insensitive to the parameter b ; namely $t(S_{min}) = [(1+3b)/(1+b)^2]S_0/(2v_{imp}) \approx (0.5 - 0.56)S_0/v_{imp}$ for $0 < b < 1$.

Using Eq. (10) in Eq. (3) gives an expression for stagnation pressure,

$$P_{stag} = P_0 \left[1 + \frac{2\rho v_{imp}^2}{3P_0} \frac{\delta R_{ave}}{S_0} (1-f^2) \right]^{5/2}. \quad (11)$$

In the ICF implosion limit, $\rho v_{imp}^2 \gg P_0$, Eq. (11) can be approximated as

$$P_{stag} \approx \frac{v_{imp}^5}{P_0^{3/2}} \left[\frac{2\rho\delta R_{ave}}{3S_0} (1-f^2) \right]^{5/2}. \quad (12)$$

The degradation of P_{stag} due to the asymmetry factor, $(1-f^2)^{5/2}$, in Eq. (12) is plotted in Fig. 2. Comparing Eq. (12) with Eq. (21) of Ref. 19 or Eq. (19) of Ref. 7 suggests the identification of $\epsilon = 1 - f^2 = 1 - \Delta E_{res}/E_{sym}$, where E_{sym} is the total hot-spot internal energy of a perfectly symmetric (spherically one-dimensional, 1D) implosion at stagnation, ΔE_{res} is the “residual energy” which is the total shell energy plus hot-spot kinetic energy difference between 2D and 1D at stagnation, and ϵ is the inefficiency of converting shell kinetic energy into hot-spot internal energy.

The hot-spot internal energy at stagnation, $E_{hs} = \frac{3}{2}P_{stag}V_{hs}$, can be obtained from Eq. (11) and Eq. (10)

$$E_{hs} = \frac{3}{2}P_0AS_0 \left[1 + \frac{2\rho v_{imp}^2}{3P_0} \frac{\delta R_{ave}}{S_0} (1-f^2) \right]. \quad (13)$$

Again expanding in the ICF limit of $\rho v_{imp}^2 \gg P_0$, Eq. (13) reduces to

$$E_{hs} \approx A\rho\delta R_{ave}v_{imp}^2(1-f^2). \quad (14)$$

The degradation of E_{hs} with mode-1 asymmetry, $1-f^2$, is shown graphically in Fig. 2. Equation (14) confirms the linkage above between f and ΔE_{res} for a mode-1 asymmetry, namely that $\Delta E_{res} = \frac{1}{2}m_{shell}v_{imp}^2f^2 = RKE$, with $m_{shell} = 2\rho\delta R_{ave}A$. In the context of the present incompressible asymmetric piston model, the mode-1 asymmetry ΔE_{res} is entirely RKE of the shell, but in a real implosion there is some RKE in the hot-spot DT and in the outer radii region of a compressible implosion that will be in-falling even as the inner radial region of the implosion stagnates upon the hot-spot. Moreover, in a real ICF implosion, some shell kinetic energy is inevitably used to compress the shell itself in addition to the hot-spot, which is an effect beyond the scope of the present analytical model but its physics is necessary to include when examining the impact of RKE of compressible simulation results or data. More discussion of RKE will follow below, but it is first necessary to note a few more model results that characterize the stagnated state.

Inertial confinement time, the characteristic existence-time of the high-pressure hot-spot, can be obtained from Eqs. (10) and (4)

$$\tau = \sqrt{\frac{S}{\dot{S}}}\bigg|_{S_{min}} = \sqrt{\frac{\rho\delta R_{ave}}{2P_0} \frac{\sqrt{S_0(1-f^2)}}{\left[1 + \frac{2\rho v_{imp}^2}{3P_0} \frac{\delta R_{ave}}{S_0} (1-f^2) \right]^2}}. \quad (15)$$

Equation (15) illustrates that asymmetry can both prolong or truncate confinement time (see Fig. 2) depending upon the degree of mode-1 asymmetry. Since the mechanical confinement-time in an ICF implosion is commonly related to burn-width, it is not unreasonable to assume, based upon Eq. (15), that observed burn-widths should be impacted by asymmetry, as has been noted in NIF data²⁰ for mode 2 asymmetry due to the desynchronization of the different elements of the implosion shell as seen in simulations.^{6,8} Examining Fig. 2, it appears that the impact on τ is slight while f is small.

If asymmetry is not too severe, such that $1-f^2 \sim \mathcal{O}(1)$ then in the limit of $\rho v_{imp}^2 \gg P_0$, it is easy to show that $\tau \approx (\sqrt{3}/2)(S_{min}/v_{imp})$.

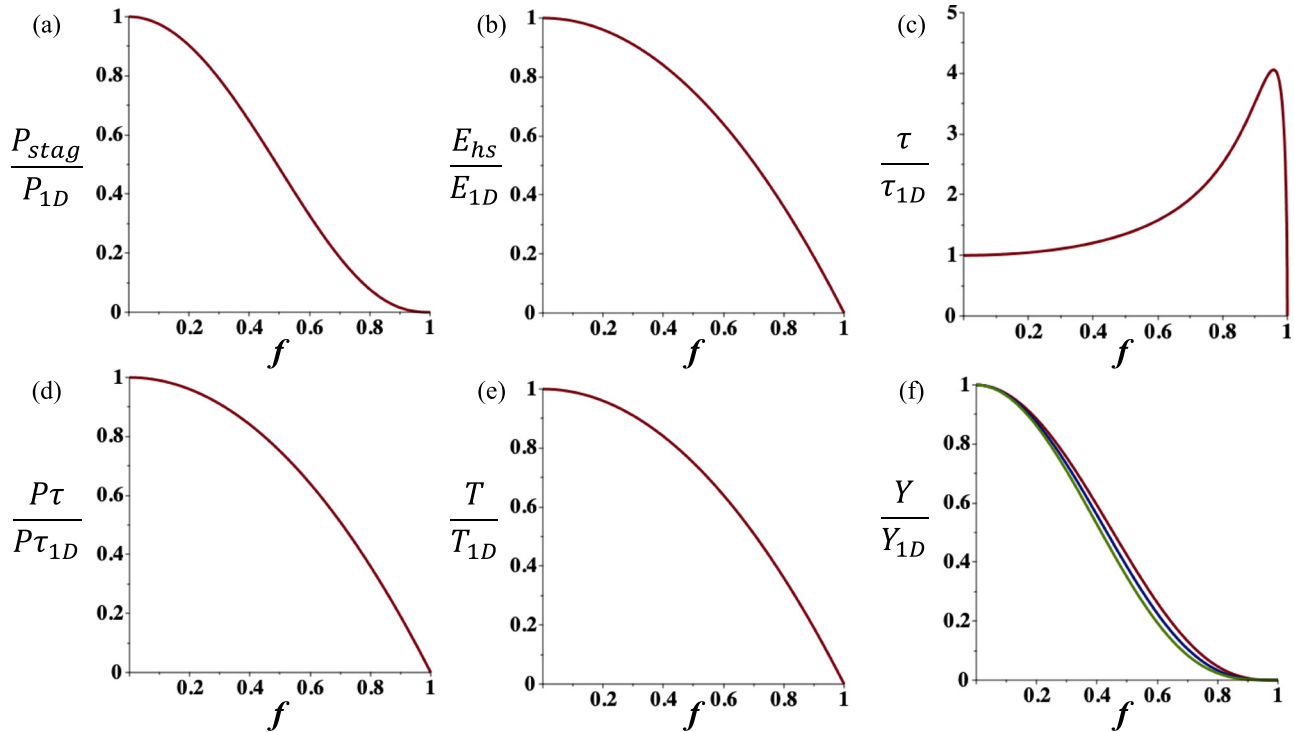


FIG. 2. Analytical model mode-1 asymmetry impact on stagnation pressure (a), hot-spot energy (b), inertial confinement-time (c), Lawson parameter (d), thermal temperature (e), and fusion yield (f). The three curves shown in (f) illustrate the small impact of assuming different reaction-rate temperature dependencies; $T^{3.8}$, $T^{3.3}$, and $T^{2.8}$ from bottom to top respectively. Here, the 1D normalizations in the ordinates are the formulas of the respective expressions, but with $f = 0$, e.g., P_{1D} corresponds to Eq. (12) with $f = 0$.

The maximum of τ , from $\partial\tau/\partial f = 0$, is the simple result $\tau_{\max} = (9/32)(S_0/v_{\text{imp}})$ and occurs when $f = \sqrt{1 - P_0 S_0 / (2\rho\delta R_{\text{ave}} v_{\text{imp}}^2)}$.

Now with expressions for stagnation pressure and confinement-time, the Lawson parameter can be constructed from Eqs. (11) and (15)

$$P_{\text{stag}}\tau = \sqrt{\frac{\rho\delta R_{\text{ave}} P_0 S_0 (1-f^2)}{2}} \times \sqrt{1 + \frac{2\rho v_{\text{imp}}^2 \delta R_{\text{ave}}}{3 P_0 S_0} (1-f^2)}. \quad (16)$$

This Lawson parameter simplifies greatly in the ICF limit $\rho v_{\text{imp}}^2 \gg P_0$ becoming

$$P_{\text{stag}}\tau \approx \rho R_{\text{ave}} (1-f^2) \frac{v_{\text{imp}}}{\sqrt{3}}, \quad (17)$$

which makes it explicit that high areal density, high velocity, and low asymmetry ($f \sim 0$) are key for moving closer to ignition conditions. Like for hot-spot energy, Eq. (14), the hot-spot gas conditions at peak velocity (P_0 and S_0) drop out of Eq. (17).

While expressions for $P_{\text{stag}}\tau$, τ , E_{hs} , and P_{stag} flow naturally from the initial assumptions of the asymmetric piston model, obtaining an expression for hot-spot thermal temperature, T_{hs} , requires an additional physics assumption. Usually, an electron-conduction limited assumption is made^{19,21} for ICF implosions to connect T_{hs} to v_{imp} by

balancing the energy loss-rate of conduction with pdV -work. Depending upon the conduction model, relations such as $T_{\text{hs}}^{7/2} \sim P_{\text{stag}} v_{\text{imp}} S_{\text{min}}$ (Spitzer²²) or $T^3 \sim P_{\text{stag}}^{4/5} v_{\text{imp}} S_{\text{min}}$ (SESAME²³) are obtained resulting in a sub (but nearly) linear velocity dependence for T_{hs} after the algebra is worked out. However, as shown in Appendix A, in the context of this piston model using an electron-conduction limited scaling for T_{hs} does not work well when compared to simulation results, so an alternate relationship will be used instead.

Hot-spot temperature at stagnation can be found from Eq. (13) using the energy relationship $E_{\text{hs}} = c_{DT} m_{\text{hs}} T_{\text{hs}}$, where c_{DT} is the DT heat capacity [$c_{DT} = 0.115 \text{ GJ/(g keV)}$, assuming 50–50 DT] and m_{hs} is hot-spot mass at stagnation. Namely,

$$T_{\text{hs}} \approx T_{\text{eff}} \left[1 + \frac{2\rho v_{\text{imp}}^2 \delta R_{\text{ave}}}{3 P_0 S_0} (1-f^2) \right], \quad (18)$$

$$\approx \frac{2}{3} \frac{T_{\text{eff}}}{P_0 S_0} (\rho R_{\text{ave}}) v_{\text{imp}}^2 (1-f^2), \quad (19)$$

where $T_{\text{eff}} = (3P_0 S_0 A) / (2c_{DT} m_{\text{hs}})$ is a parameter that may depend upon v_{imp} and $\rho\delta R_{\text{ave}}$, because of m_{hs} , but is assumed not to depend upon the asymmetry factor f . As before, the asymmetry degradation factor in Eq. (19), which by construction is the same as that of Eq. (14), is plotted in Fig. 2.

With the above expressions for hot-spot temperature, pressure, volume, and confinement-time, the formula for fusion yield, without the feedback of α -particle self-heating, can finally be obtained. Fusion

yield scales as $Y \sim P^2(\langle\sigma v\rangle/T^2)V\tau$ where $\langle\sigma v\rangle$ is the fusion reaction rate that can be approximated by a power-law in temperature, $\langle\sigma v\rangle \sim T^a$ where a can range from 2–4. The value $a = 3.3$, which is accurate between $4 < T < 7$ keV, is assumed for the plots in this paper unless otherwise stated. So, using Eqs. (19), (12), (10), and the high $\rho v_{imp}^2/P_0 \gg 1$ limit of Eq. (15) gives

$$Y \sim \left(\frac{T_{eff}}{P_0 S_0}\right)^{a-2} A(\rho \delta R_{ave})^a v_{imp}^{2a-1} (1-f^2)^a, \quad (20)$$

where again the asymmetry degradation is plotted in Fig. 2 and in Fig. 3, where it is compared to previously published simulation results. As before, relating $\epsilon = 1 - f^2 = 1 - \Delta E_{res}/E_{sym}$, Eq. (20) gives a similar impact of mode-1 asymmetry on yield than is found in Eq. (20) or (25) of Ref. 19 or Eq. (28) of Ref. 7 all of which are arrived at through

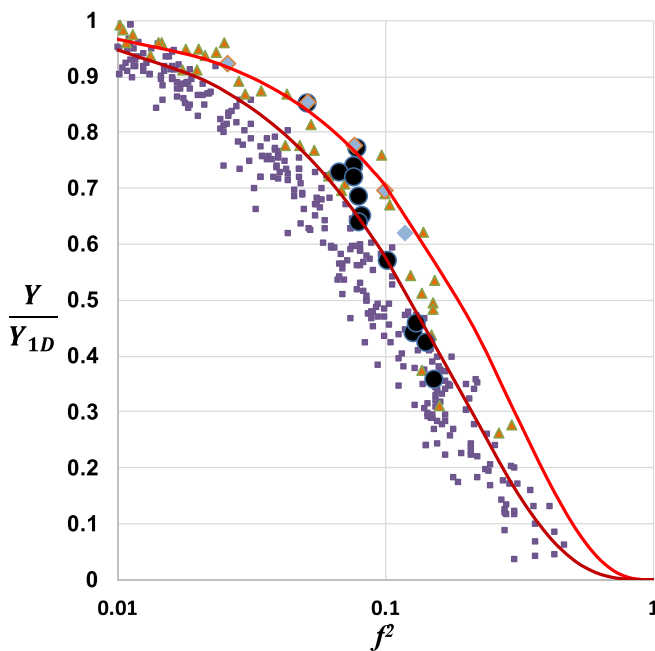


FIG. 3. The upper (red) curve is $(1-f^2)^{-3.3}$, which is the asymmetry degradation of Y shown in Eq. (20) that does not include any additional correction for alpha-particle self-heating (“no- α ”). The lower (maroon) curve includes a multiplicative correction for yield amplification effect of α -heating (“ α -on”), using Eq. (21) with a characteristic value of the parameter $\chi_{1D} = 0.88$ assumed, which is characteristic of these implosions. The orange triangles are no- α simulation data from the body of work presented in Ref. 3 (where the details of these adiabat 1.5 simulations can be found), while the purple boxes are α -on simulations from the same reference. The black circles are simulation data from Ref. 24, which are simulations of high-foot, adiabat 2.5, implosions with velocities ranging from 300 to 380 km/s that obtain temperatures of 2.5–5.5 keV. In comparing the model of Eq. (20) to the simulation data from Refs. 3 and 24, the definition $f = \sqrt{\Delta E_{res}/E_{sym}}$, suggested by Eq. (14), is used instead of Eq. (7) because the shell areal density variation was not saved in those older studies. Examination of the simulation databases from Refs. 3 and 24 indicates that for good approximation the 1D hot-spot internal energy E_{sym} is about $\sim 66\%$ of the fuel kinetic energy at peak velocity (the rest of the fuel kinetic energy largely going into cold-fuel compression) and that the change in fuel internal energy between 1D and 2D is small, so $\Delta E_{res} \approx RKE$. The blue diamonds are HYDRA²⁶ simulation results of Bigfoot^{26,27} implosions of the type described in Refs. 28 and 29 plotted against the value of f as determined from the hot-spot velocity, Eq. (23).

different physics arguments. Qualitatively similar to the mode-1 asymmetry degradation of Eq. (20), a reduction in Y due to main fuel areal density asymmetries of even modes (mode-2, 4, etc.) has also been proposed.⁴

In order to compare Eq. (20) to simulations and experiments of ICF implosion, it is also desirable to construct a multiplicative expression that can be applied to Eq. (20) to include the effect of yield amplification ($Y_{amp} = Y_{\alpha-on}/Y_{\alpha-off}$) due to α -particle self-heating. It has been shown through theoretical methods³⁰ and by fitting to simulations³¹ that Y_{amp} is related to the exponent of the product of $P\tau$ and a function T . Since expressions for the impact of mode-1 asymmetry on $P\tau$ and T are given above in Eqs. (17) and (19), the effect of mode-1 asymmetry on Y_{amp} can just be written down. Namely,

$$\frac{Y_{amp,3D}}{Y_{amp,1D}} \approx e^{\chi_{1D}^2[(1-f^2)^{a-1}-1]} \approx e^{-(a-1)\chi_{1D}^2 f^2} \quad (21)$$

using the $Y_{amp} \approx \exp(\chi_{\alpha}^{1.2})$ fit from Ref. 31, with $\chi_{\alpha} \sim P\tau\langle\sigma v\rangle/T^2 \sim P\tau T^{a-2}$ being the generalized Lawson parameter consistent with the reaction-rate assumption from above. The parameter χ_{1D} is the χ_{α} of the symmetric implosion (i.e., $f=0$). Clearly, Eq. (21) shows that yield amplification is degraded by mode-1 asymmetry because $P\tau$ and T are degraded by that asymmetry.

IV. CONNECTION OF f TO HOT-SPOT VELOCITY

It is elementary to show, based upon the abstraction shown in Fig. 1, that the velocity of the center-of-mass of the asymmetric pair of pistons is $\dot{X}_{COM} = -f v_{imp}$, where the sign indicates that the motion is directed toward the lower areal density piston. The velocity of the center of geometry of the hot-spot ($\dot{X}_{COG}$), the “hot-spot velocity,” must also be moving with the shell center-of-mass speed $v_{p1} = f v_{imp}$ toward the low areal density piston which one can reason-out geometrically or prove directly by calculating $\dot{X}_{COG} = \frac{1}{2}(\dot{x}_1 + \dot{x}_2)$ using Eqs. (1), (2), (3), and (8) obtaining

$$\dot{X}_{COG} = \frac{-f}{1-f^2} \frac{P_0 S_0^{5/3}}{\rho \delta R_{ave}} \int_{S_0}^{S_{min}} \frac{dS}{S^{5/3}} = -f v_{imp}. \quad (22)$$

Thus, we have a key result that connects shell mode-1 asymmetry to hot-spot velocity to RKE

$$f = \frac{DSR_{max} - DSR_{min}}{DSR_{max} + DSR_{min}} = \frac{v_{p1}}{v_{imp}} \approx \sqrt{\frac{RKE}{c \cdot KE}}, \quad (23)$$

where the linear proportionality between $\rho \delta R$ and the measured neutron down-scatter-ratio (DSR) has been used^{32,33} and c is a number of order unity. For the simulations shown in Fig. 3, which are representative of typical ICF implosions on NIF, $c \approx 0.66$ while for the piston model $c = 1$, suggesting that the value of c likely depends upon shell compressibility—exploring this suggestion further is a topic for another publication. Equation (23) incidentally implies that for mode-1 by measuring hot-spot p1-velocity, RKE can be inferred.

The very simple relationship shown in Eq. (23) appears to work when compared to large suites of simulations of the type described in Refs. 28 and 29 and to the majority of NIF DT implosion data (see Fig. 4). As can be seen in the right frame of Fig. 4, the asymmetric piston model does appear to predict a higher hot-spot velocity than is measured for some data around 80 km/s and it is speculated that this is the saturation of hot-spot velocity for very severe mode-1 asymmetry—a

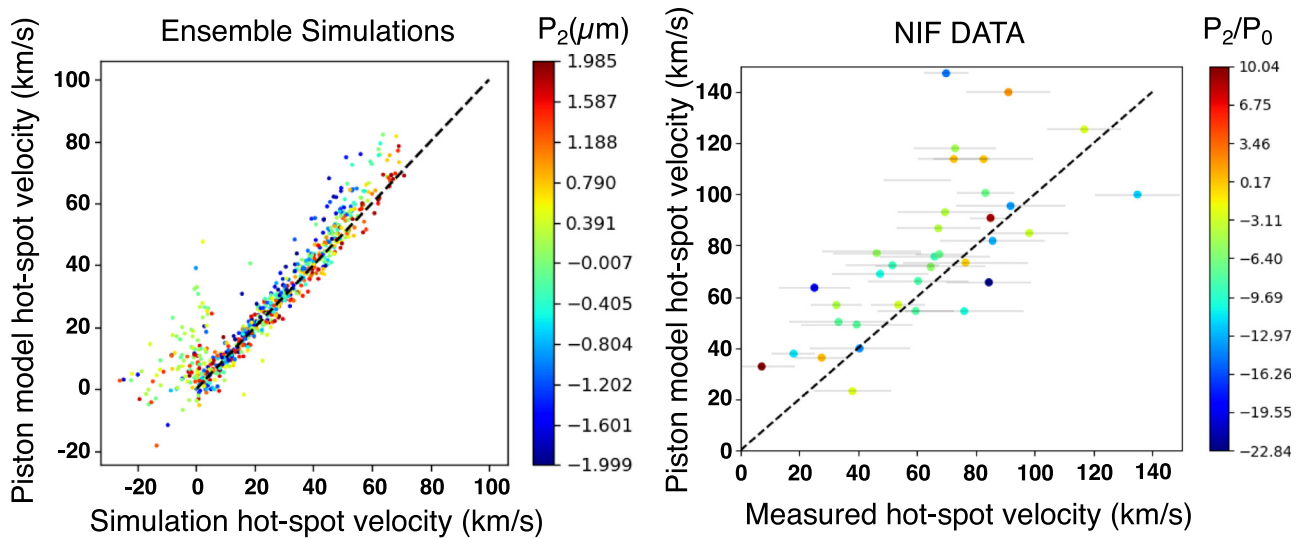


FIG. 4. Utilizing the DSR asymmetry, Eq. (23) is used to calculate the hot-spot velocity, v_{p1} and the results are compared to a large suite of simulations (left) and to NIF data (right). (Left) The simulation data are filtered for burn-average Legendre mode-2 (P_2) asymmetries less than $2\ \mu\text{m}$ and color-coded by the level of P_2 asymmetry. (Right) Experimental data are color coded by the ratio of burn-averages, P_2/P_0 , where P_0 is Legendre mode-0, i.e., hot-spot radius.

mechanism observed in published simulations of Spears.¹⁰ In Fig. 4, the hot-spot velocity in both the data and simulation is calculated from the time-integrated neutron spectra³⁴ along multiple lines-of-sight that allow one to determine the bulk burn-averaged fluid velocity in the fusing plasma.³⁵

V. COMPARISON TO HOT-SPOT PERFORMANCE DATA

It is useful to compare some of the above theory expressions of Sec. III to NIF DT implosion data. In particular, stagnation pressure and hot-spot energy will be examined. Experimentally burn-averaged stagnation pressure and hot-spot energy are inferred from the total measured fusion yield (Y_{total}), gamma reaction-history burn-width (τ_{BW}), burn-averaged temperature (T_{ion}), and hot-spot imaging data (for hot-spot volume, V_{hs}) essentially via the formula $P \sim \sqrt{T_{\text{ion}}^2 Y_{\text{total}} / \langle \sigma v \rangle V_{hs} \tau_{BW}}$ —a technique first described in Refs. 50 and 51. In the present paper, we use updated inferred data, generated with a standardized set of assumptions.³⁷ First, the appearance of the variables P_0 , S_0 , and A in the above equations must be dealt with since P_0 and S_0 historically have not been measured or inferred in DT implosion experiments.

Interestingly, by using the $\rho v_{\text{imp}}^2 \gg P_0$ limit of Eq. (10), solving for S_0 , then using that result in Eq. (12) one can eliminate both the S_0 and P_0 dependences in Eq. (12) obtaining an alternate, but equivalent, expression for hot-spot pressure in piston model geometry. Namely,

$$P_{\text{stag}} = \frac{1}{3} \frac{\rho \delta R_{\text{ave}} v_{\text{imp}}^2}{R_{hs}} (1 - f^2), \quad (24)$$

where the burn-average hot-spot radius, $R_{hs} = S_{\text{min}}/2$, has been defined. Note the difference in DSR , v_{imp} , and $(1 - f^2)$ scaling in Eq. (24) vs Eq. (12) simply results from the dependence of the same variables implicit in R_{hs} .

The $1/3$ coefficient in Eq. (24) is a result of the abstracted piston geometry of Fig. 1 and it is easy to show that this coefficient is unity

for a spheroidal geometry shell (see the Appendix B). Dropping the $1/3$ in Eq. (24) and re-writing in absolute units give another expression for hot-spot stagnation pressure, without the feedback of α -heating, that can be used to compare to real implosion data

$$P_{hs}(\text{Gbar}) = 0.0097 \frac{DSR(\%) v_{\text{imp}}^2(\text{km/s})}{R_{hs}(\mu\text{m})} (1 - f^2), \quad (25)$$

where the conversion $\rho \delta R_{\text{ave}}(\text{g/cm}^2) = 19.3 DSR$ has been used.^{32,33} Given Eq. (25), the most consistent choice for the piston-model cross-sectional area A in Eq. (14) is $A \sim R_{hs}^2$.

In Fig. 5 (upper left frame), Eq. (25) is compared to ensemble simulation data, of the type described in Refs. 28 and 29, filtered for mode-2 asymmetries of less than $2\ \mu\text{m}$ Legendre-mode amplitude. As can be observed in Fig. 5, there is a very good correlation between the prediction of Eq. (25) and the burn-average pressure calculated from simulations up to inferred Y_{amp} levels of ~ 2 . For $Y_{\text{amp}} > 2$, the simulation data rise (exponentially) above the theory curve defined by Eq. (25) as expected.

Plotting Eq. (25) against a large suite of NIF DT implosion data from 2011 to early 2019 (see Fig. 5 lower left frame) reveals that the data also correlate with the theory, except for the low-foot (National Ignition Campaign) data, highlighted in blue, that fall below the curve and are known to have been heavily impacted by ablation-front driven instability and mixing.⁴⁰ In Fig. 5, the high-foot^{20,41–46} and high-density carbon (HDC)^{47–49} are also highlighted and can be seen to correlate with the theory expression, Eq. (25), for lower hot-spot pressures but then rise above the theory curve for higher pressures, consistent with what is expected for levels of α -heating with $Y_{\text{amp}} > 2$. Note that the very highest pressure (~ 280 Gbar) high-foot datum that stands out above the other high-foot points in Fig. 5 is the result of a target engineering change that involves an alternate capsule mounting technique.⁵²

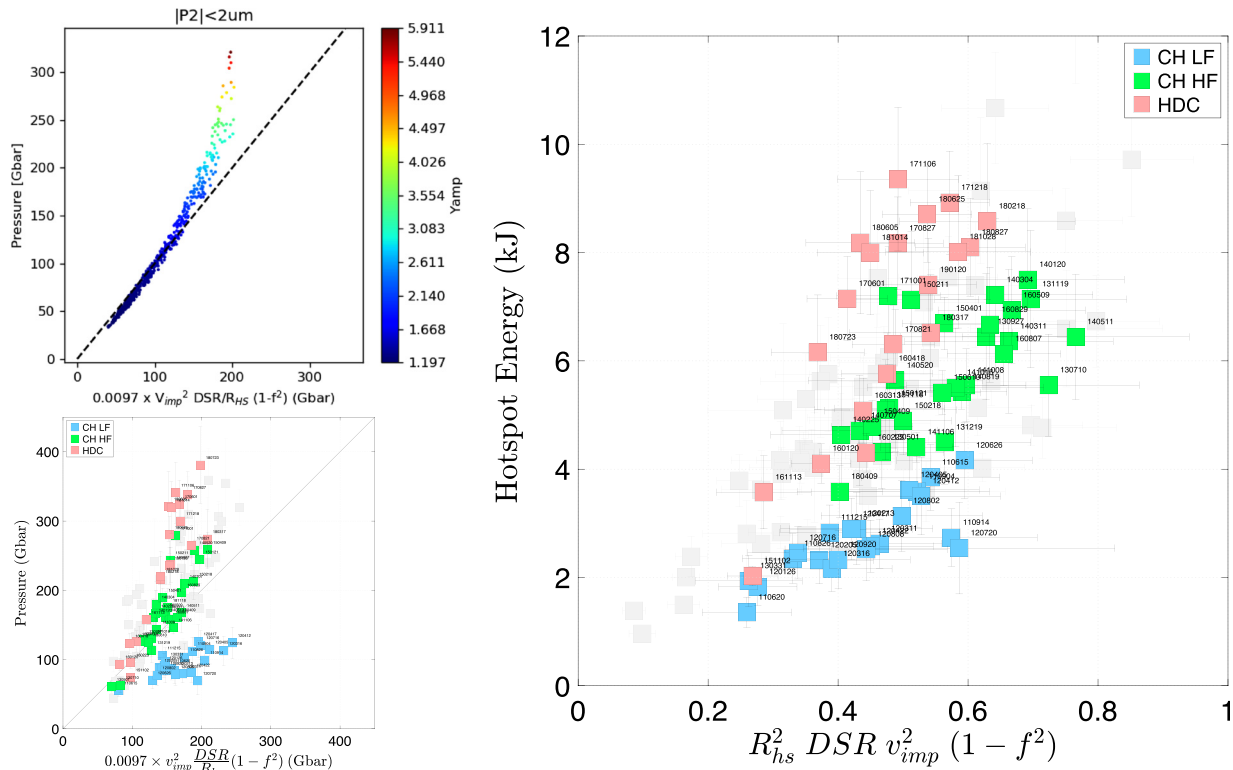


FIG. 5. Shown in the upper left frame is HYDRA²⁵ ensemble simulation derived hot-spot pressure data filtered for mode-2 asymmetries (P_2) less than $2\ \mu\text{m}$, plotted against the theory expression for hot-spot pressure and colored by Y_{amp} as inferred from the simulations using ITFX.³⁶ As expected as alpha-heating levels increase, leading to higher levels of Y_{amp} , the simulation data rise above the no- α -heating theory curve. Also shown are data from all NIF DT shots plotted against the theory expressions for hot-spot pressure (lower left frame) and energy, Eq. (14) (right frame). These inferred hot-spot pressure and energy data come from a recent re-evaluation³⁷ that put all the data accumulated over the years onto a consistent basis. While most DT shot series are shown in gray, a few key series are highlighted in color to illustrate the trending of those series—trends which are obscured if all series were colored. The key series highlighted are the low-foot (National Ignition Campaign) implosions^{38,39} (CH LF in blue) which are known to have been heavily impacted by instability-driven mixing,⁴⁰ high-foot implosions^{20,41–46} (CH HF in green), and high density carbon (“diamond” ablator) implosions^{47–49} (HDC in salmon). On these plots, there is little practical difference between HDC data and that from Bigfoot implosions,^{26,27} which comprise some of the grayed-out points. In this figure, f is calculated from the measured DSR asymmetry.

Similar to the stagnation pressure plot, the right side of Fig. 5 shows the inferred hot-spot energy plotted against Eq. (14) (again with $A \sim R_{\text{hs}}^2$) for all NIF DT implosions from 2011 to early 2019. In the hot-spot energy plot, the trend of improvement from going from low-foot implosions, to high-foot implosions, to HDC implosions is clear. Higher levels of inferred hot-spot energy for a given value of Eq. (14) imply a higher efficiency implosion.

VI. SPEC ON ALLOWABLE MODE-1 ASYMMETRY

Having identified the majority of the sources of mode-1 asymmetry,⁵³ this model of this paper can be used to set an engineering spec on the allowable asymmetry fraction given a desired tolerance of fusion yield variation ($\Delta Y/Y_{\text{ID}}$). Using Eqs. (20) and (21), one can write down a transcendental equation for the tolerance on f . Namely,

$$1 - \frac{\Delta Y}{Y_{\text{ID}}} = (1 - f^2)^a e^{-(a-1)\chi_{\text{ID}}^2 f^2}, \quad (26)$$

where Y_{ID} is defined as Eq. (20) with $f=0$.

Figure 6 illustrates the solution of Eq. (26) with $a=3.3$. As can be seen in Fig. 6, the tolerance on f gets more restrictive for higher values of χ_x , which mathematically expresses the intuitive result that in order to approach ignition the requirements on precision control of the implosion must become tighter. It should be pointed out that simulation studies often find reduced sensitivity to asymmetry for robustly igniting implosions,^{3,54} but it is a mistake to mix yield sensitivity of an implosion that is robustly igniting with the ability of that implosion to get to ignition.

Table I gives the maximum f allowed for a given tolerance on yield degradation that is required for ignition as determined from Eq. (26). For a 5% tolerance on yield degradation near ignition ($\chi_x \sim 2$), giving some allowance for other forms of yield degradation (e.g., mix) that are beyond the scope of this paper, the asymmetry fraction must be limited to $f < 0.08$. In real terms, this result means that all the engineering contributors to implosion asymmetry (laser beam-to-beam imbalances, as-built capsule thickness variations, hohlraum diagnostic window losses, etc.) must all be limited so that their additive effect results in no more than a

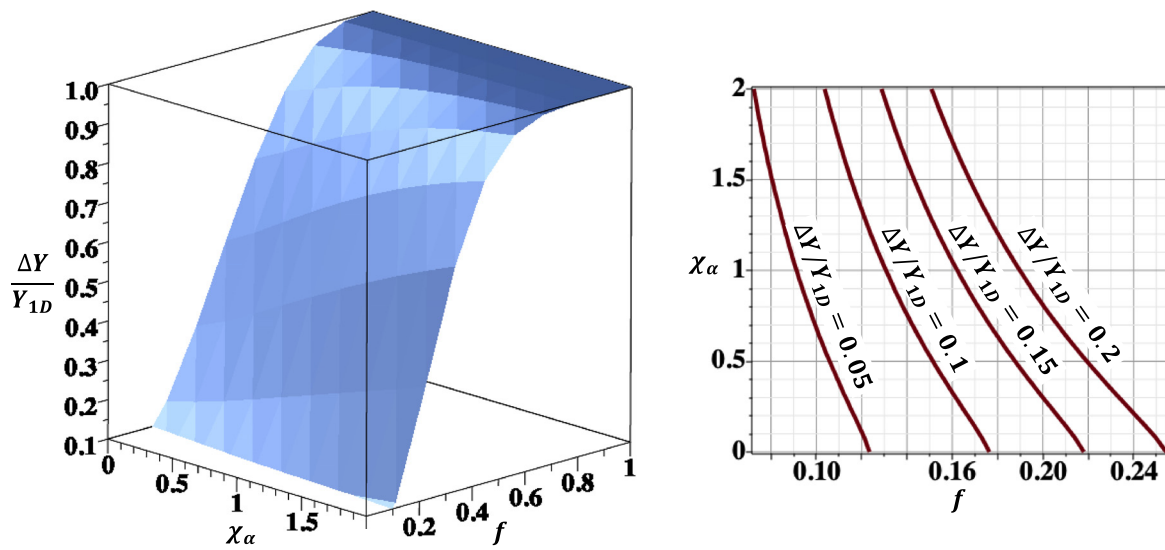


FIG. 6. (Left) The three-dimensional surface represented by the solution of Eq. (26). (Right) For arbitrarily chosen, but reasonable, sets of values of $\Delta Y/Y_{1D} = 0.05$ – 0.2 , the solution of Eq. (26) gives curves that illustrate how the tolerance on f shrinks as ignition is approached (i.e., χ_α gets larger).

30 km/s hot-spot drift velocity for a characteristic 380 km/s ICF implosion.

VII. CONCLUSION

The key conclusion from this paper is that for an ICF implosion the degradation of hot-spot performance metrics, due to shell asymmetry, scales as $(1 - f^2)^n$, where $n = 5/2$ for stagnation pressure, $n = 1$ for hot-spot energy, and Lawson parameter ($p\tau$) when the effects of alpha-heating are neglected. For thermal temperature, $n = 1$ when an adiabatic assumption applies, n can be much smaller under conditions where an electron conduction limited assumption applies (see Appendix A). The degradation of fusion yield (without alpha-heating) scales as $(1 - f^2)^a$, when the fusion reaction-rate scales with temperature as $\langle \sigma v \rangle \sim T^a$. When alpha-heating is present, the yield amplification degrades exponentially [see Eq. (21)] as a result of asymmetry. The shell asymmetry fraction, f , given by Eq. (23), directly relates mode-1 shell asymmetry alternative measures of asymmetry such as mode-1 hot-spot velocity and the concept of residual kinetic energy. Equation (25) provides a useful expression for stagnation pressure that can be directly compared to experiment. These results are based upon an asymmetric piston model abstraction of an ICF implosion with mode-1 asymmetry, without speculating the cause of the asymmetry. The key element of the mode-1 asymmetry is asymmetry in the areal density, not the hot-spot shape, and not the shell shape *per se*.

TABLE I. Spec on f with $a = 3.3$.

$\Delta Y/Y$	Max. f	Max. v_{p1} (km/s)
0.05	0.08	30
0.1	0.11	40
0.15	0.14	50
0.2	0.16	60

The model presented here works surprisingly well when compared to ensembles of high-fidelity simulations and when compared to data, but the real power of the model is as an aid to understanding the essential mechanics of an asymmetric implosion. Moreover, the comparison to simulations shown in Fig. 3, where simulations included modes 2 and 4 in addition to mode 1, suggests that the mode-1 formulas may work to describe higher modes of asymmetry when that asymmetry is measured by RKE—a topic to be explored further in a future paper.

Of course, some fidelity improving extensions of the model may be merited. A full dynamic hot-spot model could be used in place of the adiabatic assumption in order to include the effects of alpha-heating explicitly in the solution, albeit Fig. 3 suggests little practical improvement would be gained from such an addition. The incompressible piston assumption could be relaxed in order to account for the finite shock-transit time across the thickness of the piston as the piston decelerates. Mass-transfer between pistons to capture angular mass-flow at stagnation could be added to account for the expected shell morphology evolution that is expected around stagnation. Finally, the effects of 3D convergence are another reasonable extension of this model. Any of the above-mentioned improvements to better emulate a real implosion would likely result in the need for a numerical solution and would very closely resemble the numerical model of Refs. 8 and 55 and lose some of the piston-model's clarity, so whether the extra complexity is worth the likely reduction in the utility of having simple analytical expressions would need to be evaluated. Appendix B and the discussion surrounding Eq. (25) suggest why the simple model presented in this paper appears to work so well when compared to detailed ICF implosion simulations and experimental data. The good agreement between this theory, simulation, and data implies that the physics of mode-1 asymmetry in ICF implosions is largely a problem of classical mechanics.

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APPENDIX A: MODEL WITH ELECTRON-CONDUCTION LIMITED T

For the interested reader and for completeness, results for hot-spot temperature and yield using the assumption of electron-conduction limited temperature that could be used to replace Eqs. (19)

and (20) are given here. As will be shown below, assuming that electron conduction limited temperature works less well as compared to the simulation results of Fig. 3 than using the results of Eqs. (19) and (20); one can speculate that this limitation is a result of the geometry of the piston-model abstraction of Fig. 1 where the pistons only cover a small section of hot-spot solid angle unlike a spherical implosion, but proving this speculation is beyond the scope of this paper.

By balancing the rate at which pdV work is done on the hot-spot by the pistons against the loss of heat in the hot-spot via electron thermal conduction to the pistons, one finds for the abstracted (right hand side) geometry of Fig. 1 the hot-spot temperature scaling formulas $T_{\text{Spitzer}}^{7/2} \sim P_{\text{stag}} v_{\text{imp}} S_{\text{min}}$ or $T_{\text{Sesame}}^3 \sim P_{\text{stag}}^{4/5} v_{\text{imp}} S_{\text{min}}$ depending upon the choice of conduction model, Spitzer²² or SESAME,²³ respectively. Then by using Eqs. (12) and (10) and doing some algebra, one obtains

$$T_{\text{Spitzer}} \sim DSR^{2/7} v_{\text{imp}}^{6/7} (1 - f^2)^{2/7} \quad \text{or} \quad (\text{A1})$$

$$T_{\text{Sesame}} \sim P_0^{1/10} S_0^{1/6} DSR^{1/6} v_{\text{imp}}^{2/3} (1 - f^2)^{1/6}, \quad (\text{A2})$$

which have a notably weaker dependence upon DSR , v_{imp} , and $(1 - f^2)$ than the relation shown in Eq. (19).

Using Eqs. (A1) or (A2), along with Eqs. (12), (10), and (15) in the yield scaling $Y \sim P^2 (\langle \sigma v \rangle / T^2) A S_{\text{min}} \tau$ where $\langle \sigma v \rangle$ with the fusion reaction rate approximated by a power-law in temperature, $\langle \sigma v \rangle \sim T^a$ like in Sec. III, one obtains after some algebra

$$Y_{\text{Spitzer}} / Y_{1D} \sim (1 - f^2)^{(10+2a)/7} \quad \text{or} \quad (\text{A3})$$

$$Y_{\text{Sesame}} / Y_{1D} \sim (1 - f^2)^{(10+a)/6}. \quad (\text{A4})$$

Equations (A3) and (A4) are compared to the simulation data and curves of Fig. 3 in Fig. 7. As can be observed in Fig. 7, the difference between Eqs. (A3) and (A4) are not practically different and the

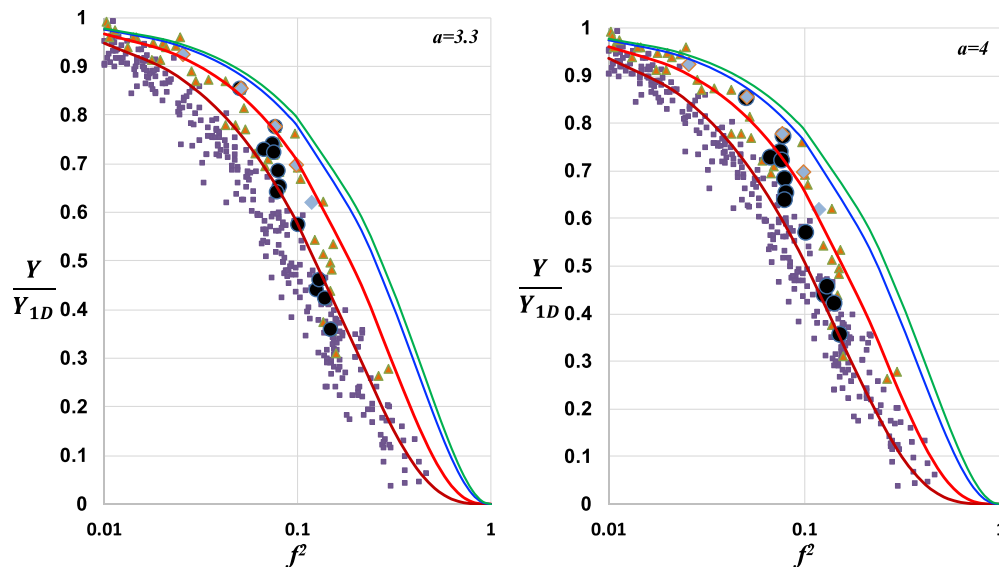


FIG. 7. Similar to Fig. 3, the degradation of yield due to asymmetry is plotted against the f^2 measure of asymmetry. The left frame assumes $a = 3.3$ in the theory curves while the right frame assumes $a = 4$. The green curves are generated using Eq. (A4) while the blue curves are generated using Eq. (A3), the other curves and data are as described in the caption for Fig. 3.

difference of either as compared to Eq. (20) is about equal to the alpha-heating correction applied to Eq. (20). For reasonable choices of a , neither Eqs. (A3) nor (A4) compare as well to the simulation results as Eq. (20).

APPENDIX B: PISTON VS SPHERICAL GEOMETRY

This section provides an elementary argument as to why the piston model geometry works for spheroidal geometry implosions. Equation (14), the hot-spot energy expression, can be re-written as

$$E_{hs} = \frac{3}{2} P_{stag} V_{hs} = \frac{1}{2} m_{shell} v_{imp}^2 (1 - f^2). \quad (B1)$$

In the piston model geometry of Fig. 1, the hot-spot volume is $V_{hs} = A S_{min} = A(2R_{hs})$ and shell mass is $m_{shell} = 2\rho\delta R_{ave}A$ which, when substituted back into Eq. (B1), gives Eq. (24) with the 1/3 coefficient out in front of the expression. For a spherical geometry hot-spot, $V_{hs} = 4\pi R_{hs}^3/3$ (for a spherical geometry hot-spot, this expression defines R_{hs}) and $m_{shell} \approx 4\pi R_{hs}^2 \rho\delta R_{ave}$ which, when substituted back in Eq. (B1), gives

$$P_{hs} = \frac{\rho\delta R_{ave} v_{imp}^2}{R_{hs}} (1 - f^2), \quad (B2)$$

which has the same $\rho\delta R_{ave}$, v_{imp} , R_{hs} , and $(1 - f^2)$ scaling dependence as Eq. (24), but the 1/3 coefficient has been canceled as a result of the geometry difference. Extrapolating from this result, it is expected that the piston model expressions for E_{hs} , τ , P_{stag} , T_{hs} , and Y can be applied as scalings to real implosions.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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